SYOFYAN ALRISANDI

+6285365103947 | syofyanalrisandi12@gmail.com | www.linkedin.com/in/syofyan-alrisandi-314381313/
Padang City, West Sumatera

Bachelor of Mechanical Engineering graduate from Andalas University in 2024 with an understanding of mechanical engineering principles, especially in design and optimization engineering. Experienced in using Autodesk Inventor, AutoCAD, SolidWorks, and MSC Nastran/Patran for CAD model development, simulation, and prototype testing. Analytical problem solver, skilled in working in a team, and committed to continuous learning. Looking to utilize skills in a mechanical engineering role or other relevant position.

Work Experiences

Badan Riset dan Inovasi Nasional (BRIN) - **Serpong, Indonesia** Jan 2023 - Feb 2023

Intern in Aerodynamics, Aeroelastics, and Aeroacoustics Laboratory

- Participate in Research and Development in the field of UAV: (1) UAV construction analysis with FEA applications; (2) Analysis of UAV fluid characteristics in the Indonesian Low Speed Tunnel (ILST)
- Analysis of the construction of a 2-engine, 19-passenger N219 Amphibious aircraft for wing components, hull connectors (Floaters), and aircraft hull with a safety factor level above 2 so that the prototype is 100% safe during testing
- Participate in aerodynamic testing of the N219 Amphibious aircraft model in the Indonesian Low Speed Tunnel (ILST)

Laboratorium Dinamika Struktur (LDS) - **Andalas University** Jan 2022 - Aug 2024

Assistant LDS Department of Mechanical Engineering

- Provided support to 153 students in college assignments related to the analysis of material strength mechanics, particle dynamics, mechanical vibrations and structural statics
- Developed solutions for homework questions and provided additional guidance
- Became an assistant and was 100% responsible for the use of laboratory equipment for the Final Assignment
- Increased student success in completing homework and Final Semester Exams by 45% by providing a more effective and clear problem-solving approach
- Structural analysis, engineering problem solving, teaching technical concepts, and effective communication

Education Level

Andalas University - **Jl. Dr. Moh. Hatta No.75, Limau Manis, Pauh District, Padang City, West Sumatra, Indonesia** Aug 2019 - Aug 2024

Degree in Mechanical Engineering, 3.61/4.00

- Final Project: Experimental Study of Natural Frequency Shift to Increase Stiffness in the Wings of a Skywalker Type Fixed Wing Unmanned Aircraft
- Learning how to make products starting from design creation, machining process planning, machine tool operation, to product quality measurement and maintenance of machine tools that have been used.
- Relevant Coursework: Technical Drawing, Engineering Materials, Manufacturing Engineering, Industrial Metrology, Mechanic of Materials, Sound Control in Industry, Mechanical Vibration, Vibration Engineering, Micro-Electrical Mechanical System, Machine Element, and Kinematic and Dynamic of Machinery.

Organisational Experience

Andalas Naval Research Development (ANRD) - **Andalas University** Apr 2023 - Dec 2023

Design and Production Division of Autonomous Tourism Surface Vessel (ATSV)

- Design and develop efficient ship designs for autonomous missions
- Produce and quality control ship prototypes to ensure the accuracy of design implementation and autonomous system integration reaches 95%.
- Conduct performance testing and simulation of autonomous missions, increasing ship operational efficiency by 95%.
- Document the design, development, and production processes for technical references for reporting needs

Andalas Flying Robot Generation (AFRG) - **Andalas University** Jun 2021 - Jan 2023

Andalas Flying Robot Generation (AFRG) Design Coordinator (starting December 2022) and Design Member for Racing Plane missions (starting July 2021)

- (Coordinator) Coordinate and support 6 design members in achieving 90% success of research and development of Unmanned Aircraft (UAV) missions; (1) Racing Plane; (2) Fixed Wing Mapping Monitoring; (3) Vertical Takeoff and Landing (4) Technology Development Proppeller (5) Technology Development Airframe (6) Technology Development Prime Mover
- (Koordinator) Providing leveling or basic knowledge of aircraft design science to 5 new members of the design division
- (Member) Designing and developing unmanned aircraft designs with a focus on the Racing Plane mission on aerodynamics and high stability aspects
- (Member) Production quality control for design implementation up to 95%.

- (Member) Conducting performance analysis trials to see the accuracy of the central of gravity and the maneuverability of the vehicle
- (Member) Documenting the design and production process for reporting and presentation purposes, accelerating the design review process
- (Member) Designing an aircraft launcher with a spring system with a success rate of 90%

Himpunan Mahasiswa Mesin (HMM FT-UNAND) - **Andalas University** May 2021 - Aug 2024

Staff of the Research and Event Division

- Team for creating 60 Mathematics Olympiad Questions and solving them at the 2021 Mechanical Creativity Competition event
- Water Rocket Judge at the 2023 Mechanical Creativity Competition event with a total of 42 participants
- Contributing to the development of the Aero-Hydro Modeling field association activity unit

Skills, Achievements & Other Experience

- **Achievements** (2023): 3rd Place in the event Lomba Desain Alat Nasional with the theme "Creative Innovation: Sustainable Wastewater Technology" by the Department of Environmental Engineering, Andalas University
- **Achievements** (2023): Finalist of the Kontes Kapal Cepat Tak Berawak Nasional (KKCTBN) division Autonomous Tourism Vessel Prototype Manufacturing and Performance by Puspresnas at Indonesia University
- **Achievements** (2023): Winner of Student Creativity Program Innovative Work (PKM-KI) Funding for the Hybrid Smart Ovitrap Team at PIMNAS 2023
- **Achievements** (2022): 1st Place in the National Energy Seminar Poster Competition by the Serikat Pekerja Pertamina Unit Pemasaran III (SPP UPms III)
- **Achievements** (2022): 2nd Place in the Lomba Desain Alat Nasional with the theme "Gen-Z Innovation: Clear Air Tech" by the Department of Environmental Engineering, Andalas University
- **Achievements** (2021): Finalist of the Kontes Robot Terbang Indonesia (KRTI), Racing Plane Division by Puspresnas
- **Achievements** (2021): 3rd Place in the Design Competition by the Andalas University Mechanical Engineering Student Association
- **Achievements** (2020): 2nd Place in the Hans-free Sanitizer Competition by the Andalas University Mechanical Engineering Student Association
- **Achievements** (2020): Hands-free Sanitizer Mehanism Design Challenge Finalist at Institution of Engineers Malaysia
- **Hard Skills**: Able to use commercial software AutoCAD (basic), Solidworks (basic), Autodesk Inventor Advanced and FEA software MSC Nastran/Patran
- **Soft Skills**: Responsibility, Teamwork, Time Management, Open-Minded, Problem Solving, Critical Thinking, Fast Learner
- **Reference**: (1) Prof. Dr. Eng, Lovely Son (+6281213552846); (2) Prof. Dr. -Ing, Mulyadi Bur (+62811661183) : Lecturer in Mechanical Engineering, Andalas University

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Final Project: Experimental Study of Natural Frequency Shift to Increase Stiffness in the Wings of a Skywalker Type Fixed Wing Unmanned Aircraft

Project Overview

- The existence of deflection that occurs in the aircraft wing construction due to the frequency of fluid disturbance approaching the value of the natural frequency of the structure (vibration resonance).
- Simulation analysis to obtain dynamic parameters in the form of structural vibration response (natural frequency, vibration mode and damping).
- Experimental analysis to obtain the actual response of the aircraft wing structure against the background of the final assignment.

Solution Method

- Modeling of the wing structure which includes the main components of the wing in the form of rip, spar, and skin using Autodesk Inventor by varying the position of the spar placement symmetrically to the central of gravity and variations in the dimensions of the spar length in the wing structure.
- Analysis of the vibration response of the structure by simulation using MSC Patran/Nastran SOL 103 for Normal Mode, the value of the natural frequency of the structure and SOL 111 to obtain a frequency response graph (FRF).
- Experimental testing of Modal Analysis to obtain a graph of the vibration response of the structure using the bump test method with an impact hummer.
- Validation of simulation and experimental data against the level of data accuracy using Mathlab.

Design Requirement

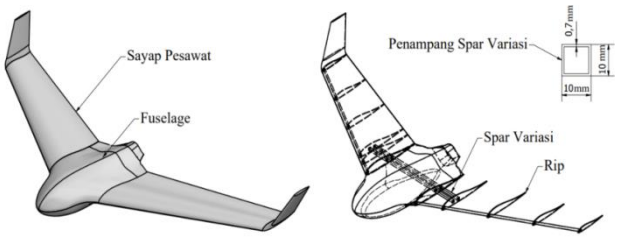


Figure 1.1 Spar use in AFRG 019 Aircraft Structure

Result

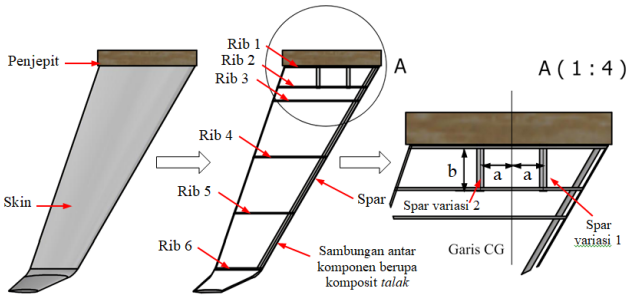


Figure 1.2 Design of the Effect of Using Spar Variations in Wing Structure

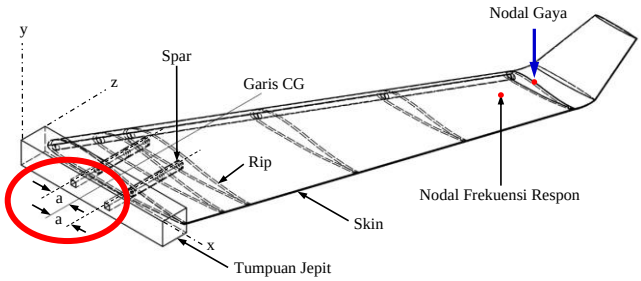


Figure 1.3 Simulation parameters of spar usage position variation

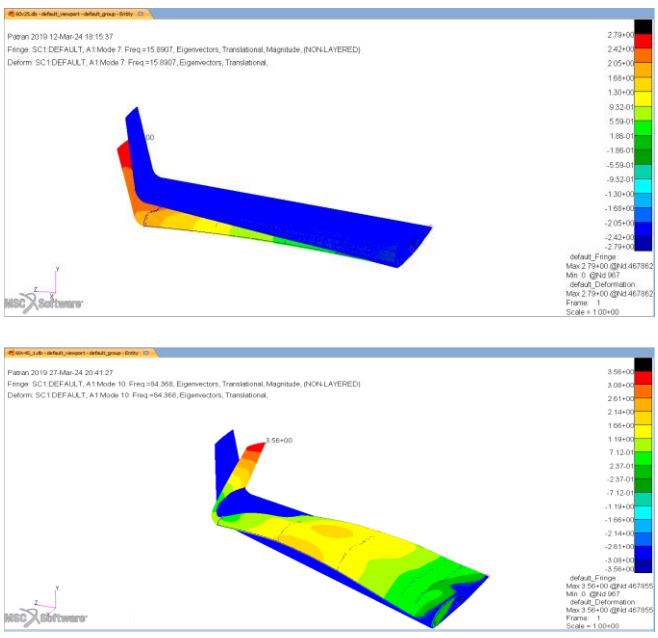


Figure 1.4 Form of Natural Frequency in Vibration Mode of Structure simulation with MSC Nastran/Patran software

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Experimental Study of Natural Frequency Shift to Increase Stiffness in the Wings of a Skywalker Type Fixed Wing Unmanned Aircraft

Continue…

Table 1.1 Value of natural frequency of 100 mm spar position variation by SOL 103 simulation

Normal Mode	Natural Frequency (Hz) Spar length 100 mm			
	Distance 15 mm from CG	Distance 25 mm from CG	Distance 35 mm from CG	Distance 45 mm from CG
1	16,3213	16,5001	16,5616	16,5437
2	42,105	49,9385	52,6888	53,2948
3	55,2678	57,3077	62,3082	66,5569
4	87,3286	90,509	94,8897	99,8773
5	121,042	124,171	127,413	130,818
6	138,571	138,713	138,965	139,416

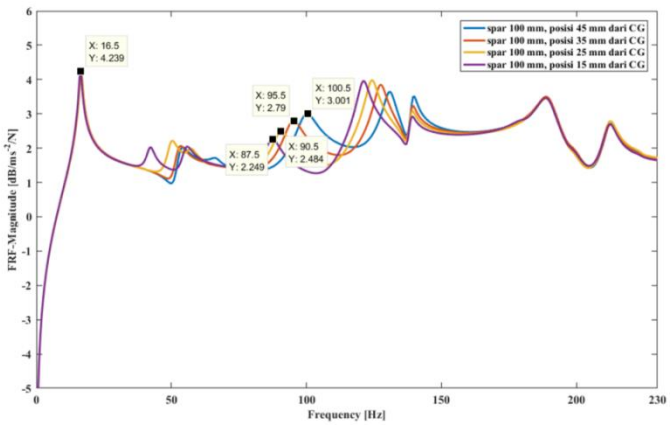


Figure 1.5 FRF of spar position variation in AFRG 019 aircraft wing structure with 100 mm spar length by SOL 111 in simulation

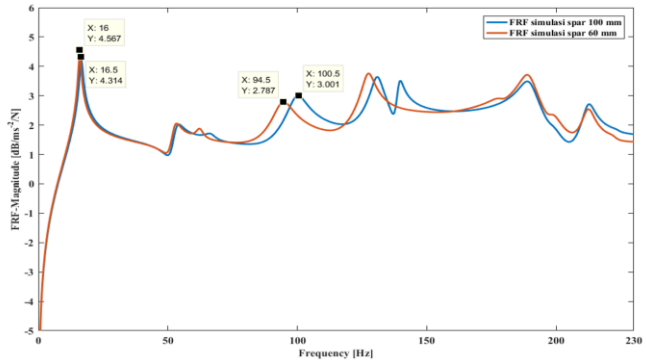


Figure 1.6 FRF Graph of Spar Length Variation in Simulation

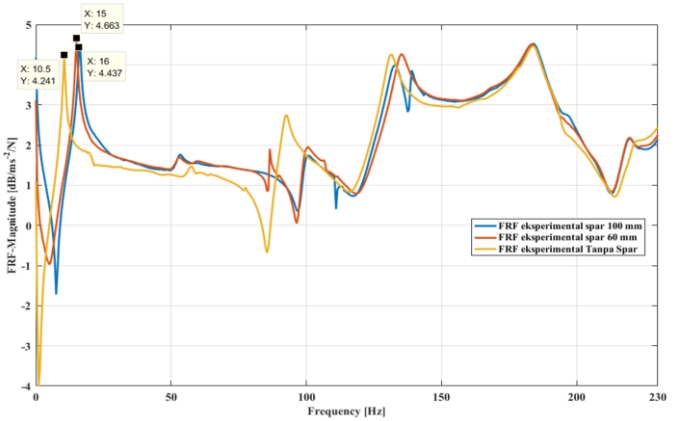


Figure 1.7 FRF Graph of Spar Length Variation in Experiment

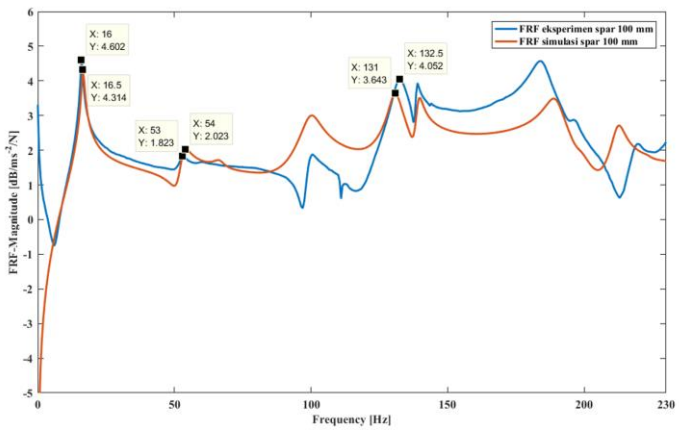


Figure 1.8 Comparison graph of FRF of 100 mm spar test and simulation

Table 1.2 Comparison of 100 mm spar length private frequency value from test and simulation

Normal Mode	Natural Frequency for Spar Length 100 mm		
	Simulation (Hz)	Experiment (Hz)	*Error* (%)
1	16,5	16,0	3,40
2	53,3	53,5	0,38
3	66,5	60,5	9,92
4	99,9	100,5	0,62
5	130,8	132,5	1,27
6	139,4	139,0	0,30

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Intership in Laboratorium Aerodinamika Aeroelatika Aeroakustika Badan Riset dan Inovasi Nasional (LA3 BRIN)

Wing Stress Analysis Study of Amphibian Aircraft Model 2 Engines with A Capacity 19 Passengers on a Scale 1:6.3 in The Indonesian Low Speed Tunnel (ILST)

Project Overview

- Aircraft construction design needs to consider the high safety factor aspect.
- Stress analysis is needed to obtain the safety factor value for safety during testing the aerodynamic parameters of the N219 aircraft.

Solution Method

- Design of the N219 Amphibious aircraft model for aerodynamic testing in the Indonesian Low Speed Tunnel (ILST) at the Aerodynamics Aeroelastics Aeroacoustics Laboratory of the National Research and Innovation Agency (LA3 BRIN).
- The aircraft design model was analyzed with a dynamic pressure of 4000 N/mm2 (at V = 80 m/s).

Design Requirement

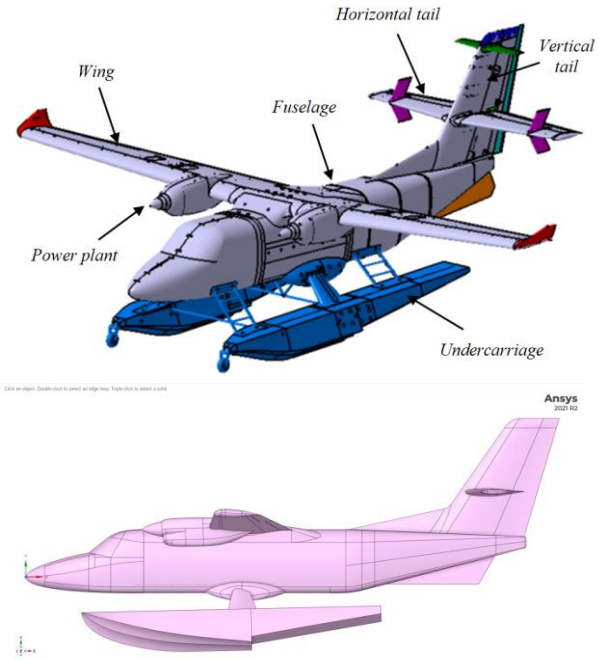


Figure 2.1 Spar use in AFRG 019 Aircraft Structure

Result

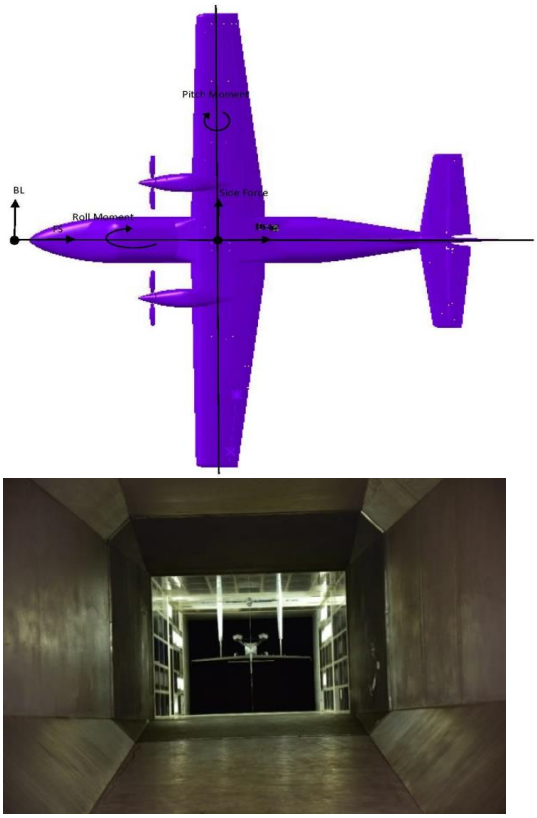


Figure 2 .2 N219 Amphibi Aircraft Wing Model

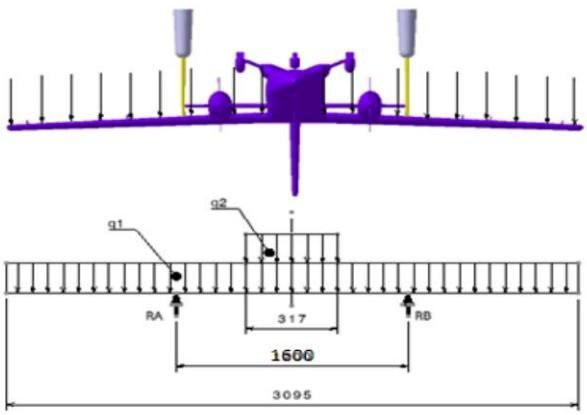


Figure 2.3 Wing strut loading distribution at normal condition

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Intership in Laboratorium Aerodinamika Aeroelatika Aeroakustika Badan Riset dan Inovasi Nasional (LA3 BRIN)

Wing Stress Analysis Study of Amphibian Aircraft Model 2 Engines with A Capacity 19 Passengers on a Scale 1:6.3 in The Indonesian Low Speed Tunnel (ILST)

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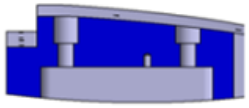
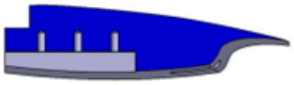
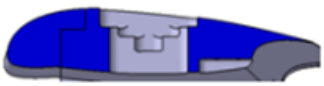
Cutting Position at the Wing (mm)	Bending MomenT (Nmm)	Normal Stress (N/mm^2)	Safety Factor
 Position 70 mm from CG (I=1.598.000 mm4, c=30 mm)	-6,906,333.30 (Blade of condition) -4,410,683.30 (Normal condition)	129.66 (Blade of condition) 82.80 (Normal condition)	3.09 (Blade of condition) 4.83 (Blade of condition)
 Position 340,75 mm from CG (I=1.598.000 mm4, c=30 mm)	-3,855,254.96	44.85	8.9
 Position 600 mm from CG (I=1.598.000 mm4, c=30 mm)	-1,802,531.51	30.69	13
 Position 671,84 mm from CG (I=1.598.000 mm4, c=30 mm)	-1,539,433.04	19.32	20.70
 Position 800 mm from CG (I=1.598.000 mm4, c=30 mm)	-1,121,564.13	23.49	17.00

Figure 2.1 Critical loading and safety factor values in the design of testing on the N219 Amphibi aircraft model at the Indonesian Low Speed Tunnel (ILST)

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Department of Mechanical Engineering, Faculty of Engineering, Andalas University: Simulation Using Autodesk Inventor Link Joint Bracket for 100 N loading

Project Overview

- Bracket is used as a support between one component and another. The bracket must have more strength related to stress analysis and safety factor in order to produce low displacement
- Analysis is carried out on a load of 100 Newton with linear analysis limits on steel material (isotropic), given a load on the joint area (pin), with the bracket perfectly connected by welding (weld connection)

Design Requirement

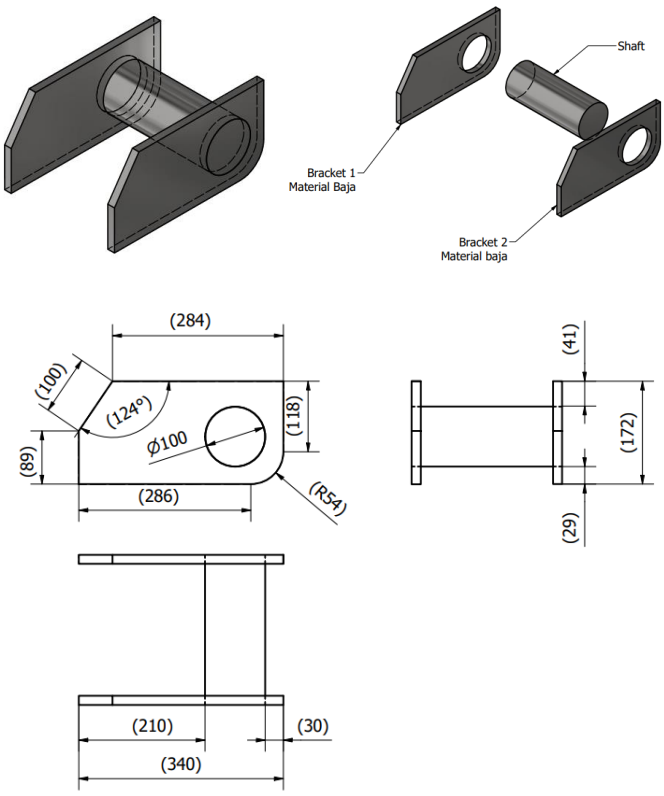


Figure 3.1 Link joint bracket design

Result

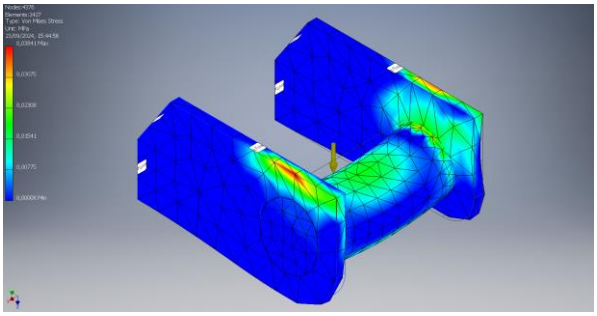


Figure 3.2 Stress analysis due to 100 N load

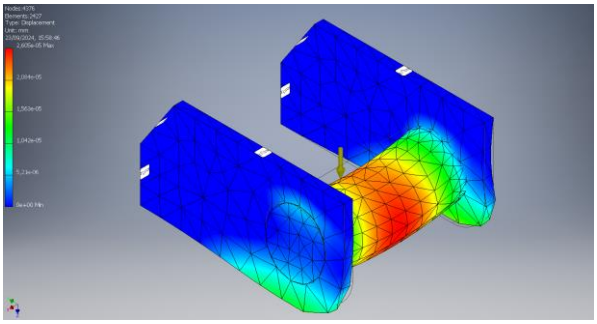


Figure 3.3 Displacement due to 100 N load

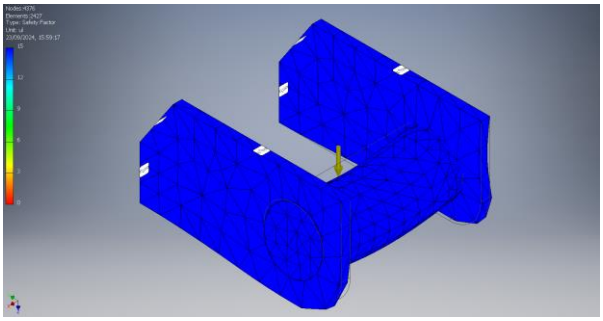


Figure 3.4 Safety factor due to 100 N load

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Andalas Flying Robot Generation (AFRG) Gonjong Tujuah-018 : Racing Plane Category of Kontes Robot Terbang Indonesia (KRTI) 2021 by Puspresnas

Project Overview

- Design of a racing plane auto with a maximum take of weight (MTOW) of 3 kg with a fast on track mission to form the number 8 at a distance of 200 meters as much as possible in 3 minutes.

Solution Method

- The wings of the aircraft are made of lightweight styrofoam that is flexible to form.
- The reinforcement is in the form of a fiber composite that has high resistance to wind fluids.
- The autonomous control system uses an ardupilot connected to a pixhawk, telemetry, receiver and GPS.

Result

- The maximum speed of the aircraft reaches 150 km/h.
- Maximum weight MTOW 2.1 kg.
- The number of tracks (number 8) formed for 3 minutes is 9 laps

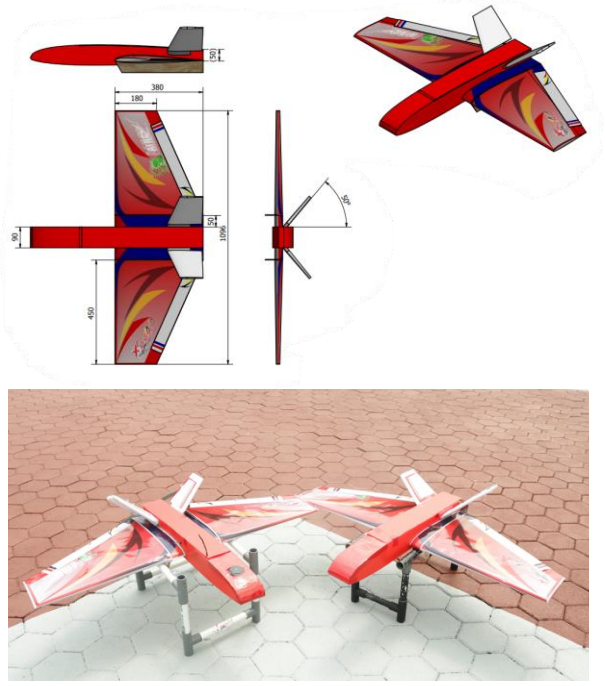
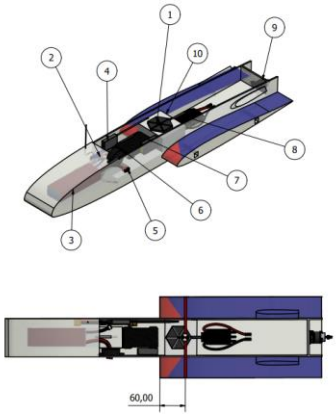
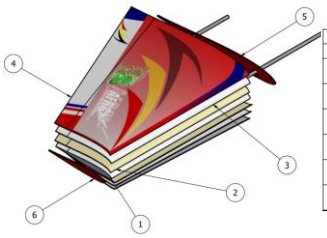


Figure 4.1 Design and Prototype Gonjong Tujuah AFRG-018



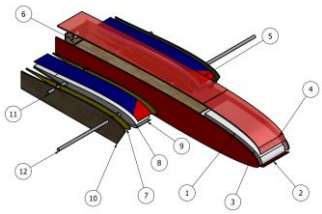
Component List	
Number	Part
1	GPS
2	Power Module
3	Batterry
4	Telemetry
5	Buzzer
6	Pixhawk
7	Receiver
8	ESC
9	Motor Brushless
10	Dudukan GPS

Figure 4.2 Wiring scheme for electrical placement



PART LIST			
NO.	Part	Material	Keterangan
1	Wing	Styrofoam	-
2	Lapisan WR	Serat WR	-
3	Lapisan Resin	Resin dan Hardener	Ketebalan 1 mm
4	Lapisan Stiker	Print Stiker	-
5	Root Wing	Impraboard	Ketebalan 3 mm
6	Wingtip	Impraboard	Ketebalan 3 mm

Figure 4.3 Components of material forming the structure of the aircraft wing



PARTS LIST			
ITEM	JUMLAH/UNIT	PART	KETERANGAN
1	1	Styrofoam Fuselage	-
2	1	Impraboard Bawah	-
3	2	Impraboard Samping	-
4	1	Lapisan Stiker Nose	Material Vnyl Stiker
5	1	Tutup Kanopi	Material Impraboard
6	1	Mouting Motor	Material Triplek 3 mm
7	4	Root Join	Material Impraboard
8	2	Styforoam Join	-
9	2	Skin Join	Material WR & Resin
10	2	Landing Gear & Stabilizer	Material Triplek
11	1	Spar Belakang Fuselage	Material Aluminium Diameter 8 mm
12	1	Spar Depan Fuselage	Material Aluminium Diameter 8 mm
13	1	Lapisan Balsa	Material Balsa Ketebalan 2 mm

Figure 4.4 Components forming the aircraft fuselage

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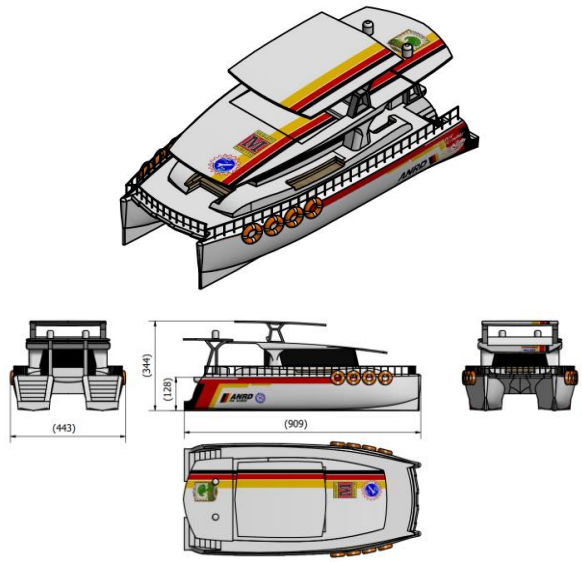
Design Engineering Andalas Naval Research Development (ANRD) RSTA A-1 Team: Manufacture and Performance of the Prototype of the Autonomous Tourism Surface Vessel (ATSV) Category of the Kontes Kapal Cepat Tak Berawak Nasional (KKCTBN) 2023 by Puspresnas at Indonesia University

Project Overview

- Designing a stable maneuvering ATSV ship with an autonomous system that is able to sense the track based on color and objects. The minimum weight of the ship is 7 kg. The ship is designed by catamaran hull with 2 hulls that are able to maneuver stably .The ship is made of fiberglass composite

Result

- The ship is able to float with a DWL of only 6 cm for a displacement of 10 kg.
- The mass of the ship is 80 kg with ship dimensions of 90x44x34 cm.
- Stable ship maneuver



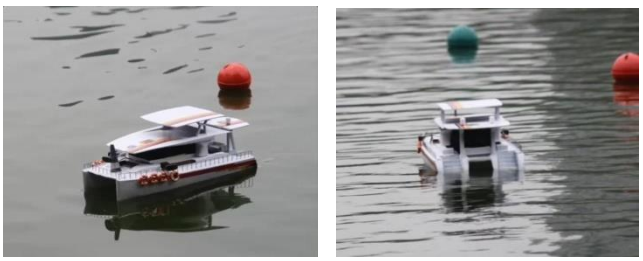


Figure 5.1 Design and Prototype RSTA A-1

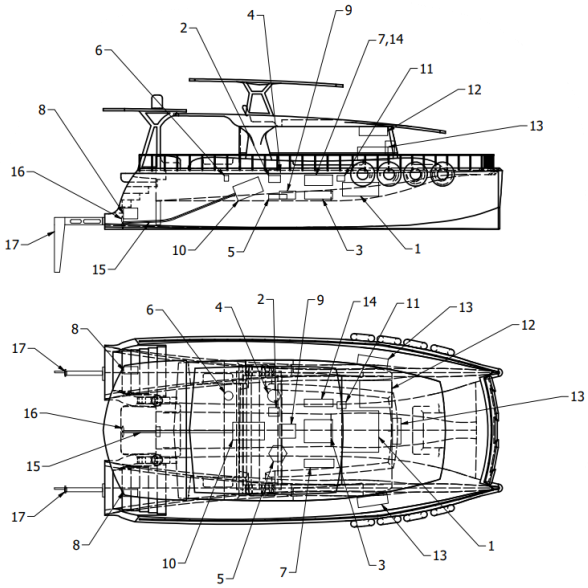


TABLE			
NO	PART LIST	JUMLAH	KETERANGAN
1	Baterai	1	Suber energi
2	Power module	1	Konneksi daya ke pixhawk
3	Pixhawk	1	Kontroler
4	Buzzer	1	Aktuator suara
5	GPS	1	Komunikasi koordinat
6	Switch	1	Pembatas arus
7	Telemetri	1	Penghubung User
8	Motor Servo	2	Aktuator Pengerak Rudder
9	ESC	1	Pengatur aiiran arus ke motor
10	Motor Brushless	1	Motor penggerak
11	Ubac	1	connector Raspberry Pi
12	Raspberry Pi	1	Kontrol Kamera
13	Kamera	3	Sensor Penginderaan
14	Receiver	1	Peangkap sinyal transmitter
15	Shaft	1	Transmisi
16	Propeller	1	Penghasil Thrust
17	Rudder	2	Manuver

Figure 5.2 Wiring scheme for electrical placement

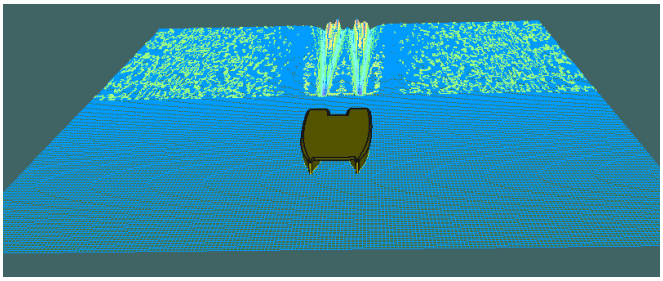


Figure 5.3 Simulation of ship design analysis RSTA A-1 on resistance and wave shape

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Pekan Kreatif Mahasiswa Karya Inovatif (PKM KI) at PIMNAS 2023: Hybrid Smart E-Ovitrap

Project Overview

- Hybrid Smart E-Ovitrap is a new breakthrough that combines Ovitrap with an IoT system for optimal mosquito eradication. There are 2 main mechanisms, electric mosquito and ovitrap (larvae killer solution). Equipped with LED mosquito lures and ovitrap solution to lure mosquitoes to lay eggs. In addition, it is also equipped with an alarm reminder to change the ovitrap solution.

Result

- All mechanisms work

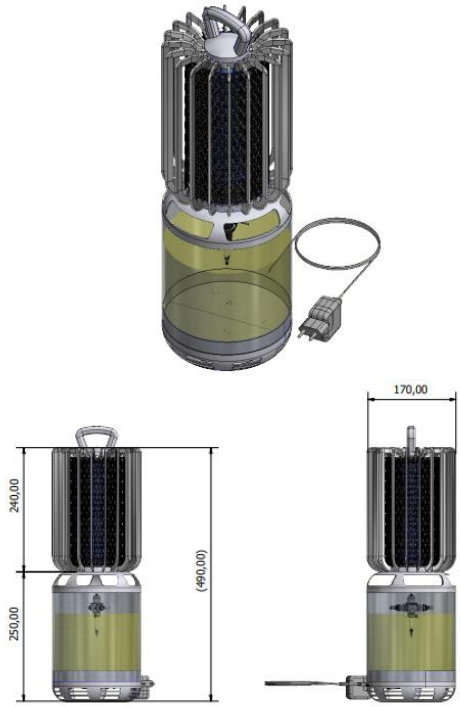




Figure 6.1 Design and Prototype Hybrid Smart E-Ovitrap

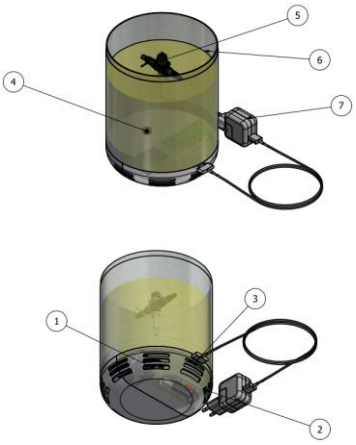


TABLE		
NO	PART LIST	KETERANGAN
1	Arduino Uno R3	Pusat Kontrol
2	Port PCB Raket Listrik	Mekanisme pembasmi nyamuk dengan aliran listrik
3	Connector USB DIP Micro	Penyambung arus masuk
4	Buzzer	alat output berupa suara
5	Turbidity Sensor	Sensor kekeruhan air akibat nyamuk
6	Larutan Ovitrap	
7	Charger	Sumber daya alat

Figure 6.2 E-Ovitrap mechanism system

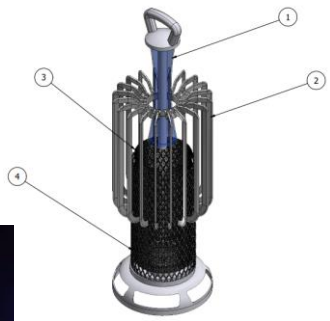


TABLE		
NO	PART LIST	KETERANGAN
1	Tabung lampu LED UV	Tempat peletakan LED
2	Penyangga	Pengaman saat kontak dengan tangan
3	Kawat Listrik Dalam	Airan listrik +
4	Kawat Listrik Luar	Aliran listrik -

Figure 6.3 Electric mosquito mechanism system

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Department of Mechanical Engineering, Faculty of Engineering, Andalas University: Automatic Fish Feed Pellet Maker Machine

Project Overview

- A machine used to make fish food pellets automatically using a coil conveyor for the pellet mixer and compactor system. Equipped with a cutting gear that moves based on a motor transmission system with a pulley belt.

Result

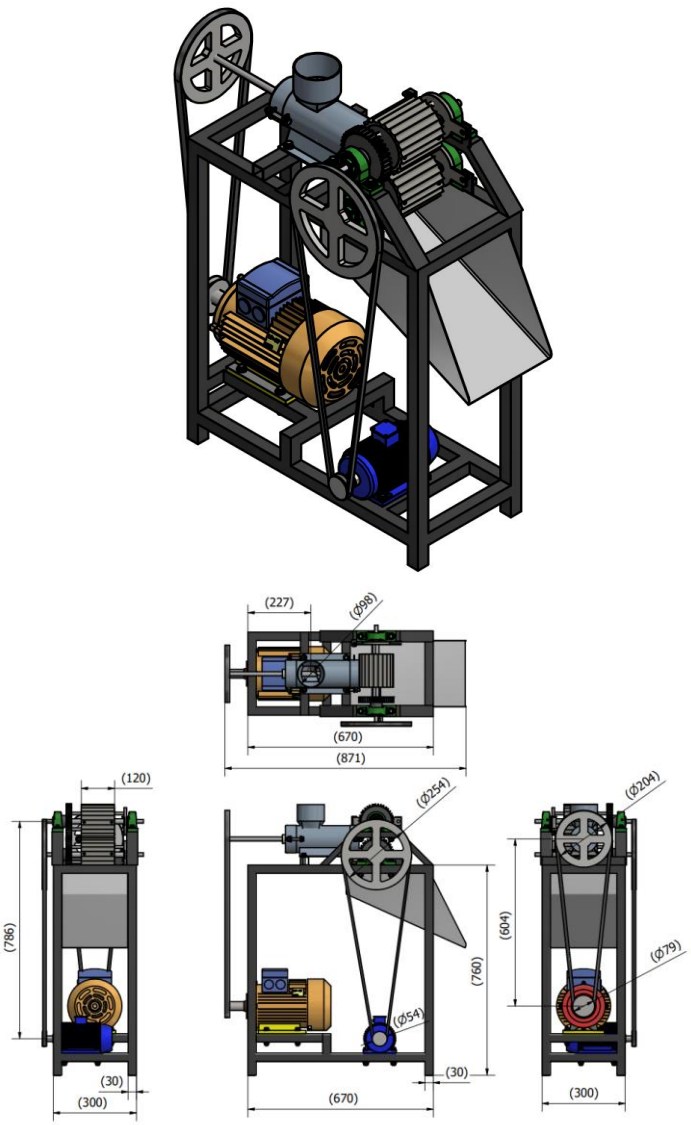
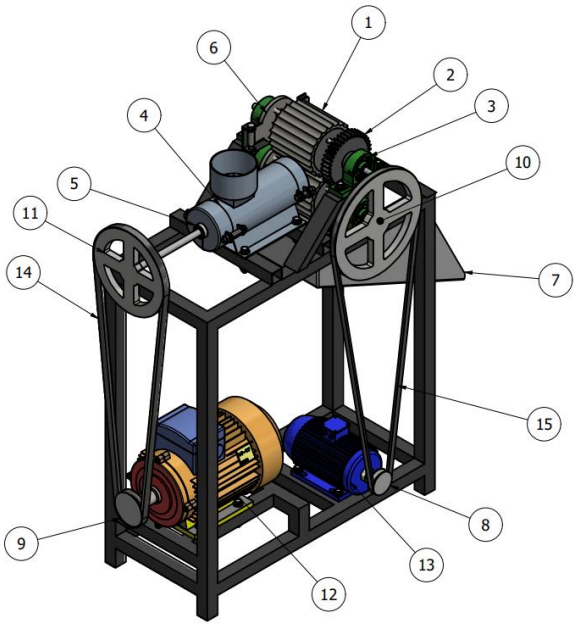


Figure 7.1 Automatic pellet maker design



PARTS LIST		
ITEM	QTY	PART NUMBER
1	2	Gear Pemotong Diameter 100 mm
2	2	Gear Tranmisi
3	4	Pillow Block Bearing as 19 mm
4	1	Screw Conveyor
5	1	Bearing
6	2	Camb
7	1	Wadah Penampung
8	1	Pulley Diameter 2 inci
9	1	Pulley Diameter 2 inci
10	1	Pulley Diameter 10 inci
11	1	Pulley Diameter 8 inci
12	1	Motor AC 1 HP
13	1	Motor AC 0,5 HP
14	1	V-belt A-64
15	1	V-belt A-68

Figure 7.2 Machine assembly components

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Department of Mechanical Engineering, Faculty of Engineering, Andalas University: Coconut Fiber Cutting Tool

Project Overview

- This machine is used for home industries for separating coconut fibers. This machine is equipped with many cutting blades to make the job easier.

Result

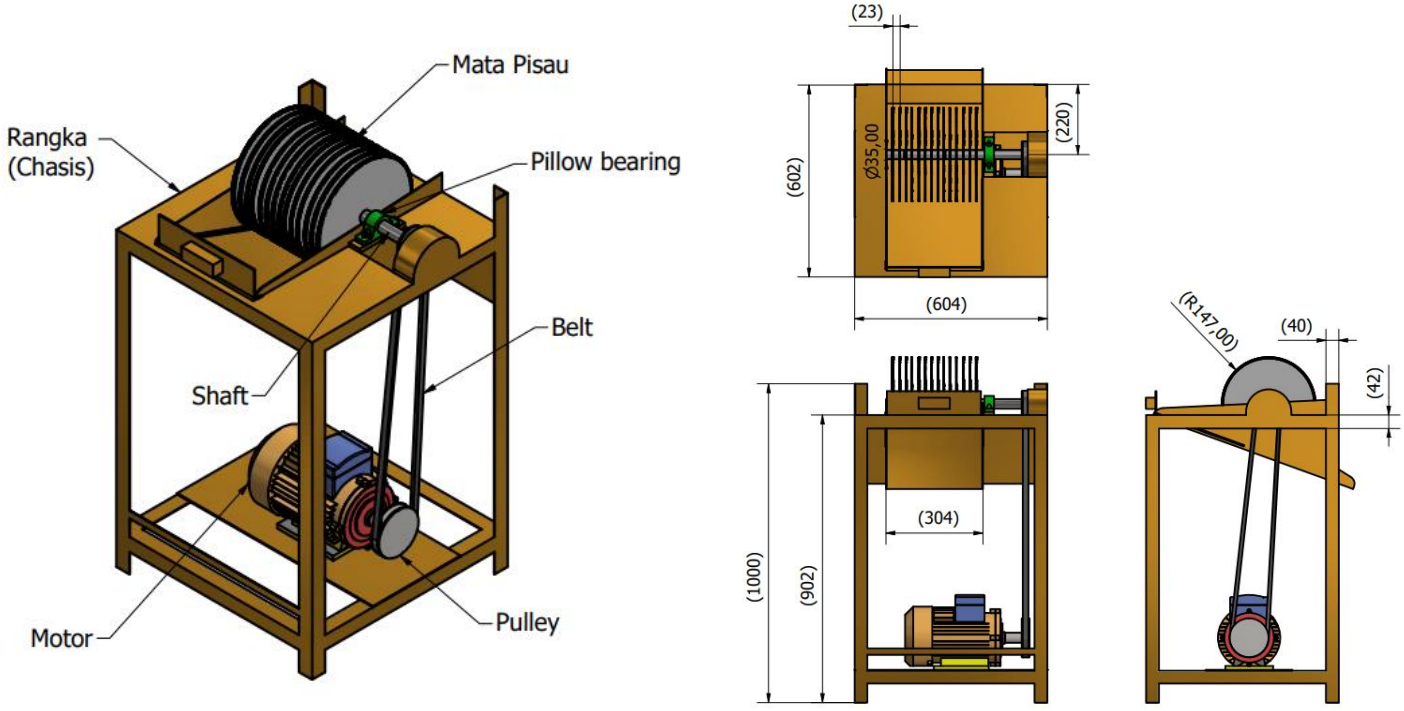


Figure 8.1 Coconut Fiber Cutting Tool design